

Frequency-domain thermoreflectance in layered medium including electron-phonon coupling

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September 17, 2025

1 The phenomenological two-temperature model

Electrons and phonons behave differently at the short-time scale, but with sufficient electron-phonon scatterings, they eventually achieve the same temperature. Specifically, the electron and phonon temperatures, T_e and T_p , evolve according to,

$$C_e \frac{\partial T_e}{\partial t} = \nabla \cdot \boldsymbol{\kappa}_e \cdot \nabla T_e - g(T_e - T_p) + P, \quad (1)$$

$$C_p \frac{\partial T_p}{\partial t} = \nabla \cdot \boldsymbol{\kappa}_p \cdot \nabla T_p + g(T_e - T_p), \quad (2)$$

where C_e and C_p are the volumetric specific heats for electrons and phonons, $\boldsymbol{\kappa}_e$ and $\boldsymbol{\kappa}_p$ are electronic and phonon thermal conductivity tensors, g is the energy transfer rate between electron and phonon subsystems per unit volume, and P is the heat source term due to pump laser heating.

In the frequency domain and reciprocal space, we have,

$$C_e i\omega \tilde{T}_e = \left(-\kappa_{e,\parallel} q_{\parallel}^2 + \kappa_{e,z} \frac{\partial^2}{\partial z^2} \right) \tilde{T}_e - g(\tilde{T}_e - \tilde{T}_p) + \tilde{P}, \quad (3)$$

$$C_p i\omega \tilde{T}_p = \left(-\kappa_{p,\parallel} q_{\parallel}^2 + \kappa_{p,z} \frac{\partial^2}{\partial z^2} \right) \tilde{T}_p + g(\tilde{T}_e - \tilde{T}_p), \quad (4)$$

where $\tilde{P}(\omega, \mathbf{q}_{\parallel}, z) = P_0 \alpha e^{-\frac{\sigma_x^2 q_x^2}{8}} e^{-\frac{\sigma_y^2 q_y^2}{8}} e^{-\alpha z}$ and P_0 is the absorbed pump power. Introducing $\Lambda_e = \frac{iC_e \omega + \kappa_{e,\parallel} q_{\parallel}^2 + g}{\kappa_{e,z}}$ and $\Lambda_p = \frac{iC_p \omega + \kappa_{p,\parallel} q_{\parallel}^2 + g}{\kappa_{p,z}}$, we rewrite Eqs. 1 and 2 as,

$$\frac{\partial^2 \tilde{T}_e}{\partial z^2} - \Lambda_e \tilde{T}_e + \frac{g}{\kappa_{e,z}} \tilde{T}_p + \frac{\tilde{P}}{\kappa_{e,z}} = 0, \quad (5)$$

$$\frac{\partial^2 \tilde{T}_p}{\partial z^2} - \Lambda_p \tilde{T}_p + \frac{g}{\kappa_{p,z}} \tilde{T}_e = 0. \quad (6)$$

Further, let us define the matrix

$$\mathbf{M} = \begin{pmatrix} \Lambda_e & -\frac{g}{\kappa_{e,z}} \\ -\frac{g}{\kappa_{p,z}} & \Lambda_p \end{pmatrix}. \quad (7)$$

Denote the eigenvalues of $\mathbf{M}$ by λ_i , and its eigenvector matrix by $\mathbf{u}$, where

$$\mathbf{u} = \begin{pmatrix} u_{11} & u_{12} \\ u_{21} & u_{22} \end{pmatrix}. \quad (8)$$

Moreover, we define the inverse of the eigenvector matrix, $\mathbf{v} = \mathbf{u}^{-1}$. Consequently, the governing equations for electron and phonon temperatures can be expressed as,

$$\frac{\partial^2}{\partial z^2} \begin{pmatrix} \tilde{T}_e \\ \tilde{T}_p \end{pmatrix} - \mathbf{M} \begin{pmatrix} \tilde{T}_e \\ \tilde{T}_p \end{pmatrix} + \begin{pmatrix} \tilde{P} \\ 0 \end{pmatrix} = 0. \quad (9)$$

By multiplying $\mathbf{v}$ on both sides, we obtain

$$\frac{\partial^2}{\partial z^2} \mathbf{v} \begin{pmatrix} \tilde{T}_e \\ \tilde{T}_p \end{pmatrix} - \begin{pmatrix} \lambda_1 & 0 \\ 0 & \lambda_2 \end{pmatrix} \mathbf{v} \begin{pmatrix} \tilde{T}_e \\ \tilde{T}_p \end{pmatrix} + \begin{pmatrix} \frac{v_{11}\tilde{P}}{\kappa_{e,z}} \\ \frac{v_{21}\tilde{P}}{\kappa_{e,z}} \end{pmatrix} = 0. \quad (10)$$

Let

$$\begin{pmatrix} \tilde{T}'_e \\ \tilde{T}'_p \end{pmatrix} = \mathbf{v} \begin{pmatrix} \tilde{T}_e \\ \tilde{T}_p \end{pmatrix}. \quad (11)$$

Specifically, the components of this new temperature vector are given by,

$$\begin{pmatrix} \tilde{T}'_e \\ \tilde{T}'_p \end{pmatrix} = \begin{pmatrix} B_1 e^{-\sqrt{\lambda_1}z} + C_1 e^{\sqrt{\lambda_1}z} + D_1 e^{-\alpha z} \\ B_2 e^{-\sqrt{\lambda_2}z} + C_2 e^{\sqrt{\lambda_2}z} + D_2 e^{-\alpha z} \end{pmatrix}. \quad (12)$$

The solution for temperatures, $\tilde{T}_e$ and $\tilde{T}_p$, can be converted back from the new variables using,

$$\begin{pmatrix} \tilde{T}_e \\ \tilde{T}_p \end{pmatrix} = \mathbf{u} \begin{pmatrix} \tilde{T}'_e \\ \tilde{T}'_p \end{pmatrix}. \quad (13)$$

Substituting Eq. 12 into Eq. 10 yields,

$$D_1 = \frac{v_{11}\tilde{P}(z=0)}{\kappa_{e,z}(\lambda_1 - \alpha^2)}, \quad (14)$$

and

$$D_2 = \frac{v_{21}\tilde{P}(z=0)}{\kappa_{e,z}(\lambda_2 - \alpha^2)}. \quad (15)$$

Applying the boundary conditions $\frac{\partial \tilde{T}_e}{\partial z} = \frac{\partial \tilde{T}_p}{\partial z} = 0$ at $z = 0$, we derive that,

$$C_1 = B_1 + \frac{\alpha}{\sqrt{\lambda_1}} D_1, \quad (16)$$

$$C_2 = B_2 + \frac{\alpha}{\sqrt{\lambda_2}} D_2. \quad (17)$$

Therefore, at $z = 0$, the coefficients B_1 and B_2 can be expressed by,

$$B_1 = \frac{u_{22}}{2U} \tilde{T}_e(0) - \frac{u_{12}}{2U} \tilde{T}_p(0) - \frac{1}{2} \left(1 + \frac{\alpha}{\sqrt{\lambda_1}} \right) D_1, \quad (18)$$

$$B_2 = -\frac{u_{21}}{2U} \tilde{T}_e(0) + \frac{u_{11}}{2U} \tilde{T}_p(0) - \frac{1}{2} \left(1 + \frac{\alpha}{\sqrt{\lambda_2}} \right) D_2, \quad (19)$$

where $U = u_{11}u_{22} - u_{12}u_{21}$.

This further suggests that electron and phonon temperatures can be expressed in terms of $\tilde{T}_e(0)$, $\tilde{T}_p(0)$, D_1 and D_2 as,

$$\begin{aligned}\tilde{T}_e &= \frac{1}{U} \left[u_{11}u_{22} \cosh \sqrt{\lambda_1}z - u_{12}u_{21} \cosh \sqrt{\lambda_2}z \right] \tilde{T}_e(0) \\ &+ \frac{u_{11}u_{12}}{U} \left[-\cosh \sqrt{\lambda_1}z + \cosh \sqrt{\lambda_2}z \right] \tilde{T}_p(0) \\ &+ u_{11} \left[\frac{\alpha}{\sqrt{\lambda_1}} \sinh \sqrt{\lambda_1}z - \cosh \sqrt{\lambda_1}z + e^{-\alpha z} \right] D_1 \\ &+ u_{12} \left[\frac{\alpha}{\sqrt{\lambda_2}} \sinh \sqrt{\lambda_2}z - \cosh \sqrt{\lambda_2}z + e^{-\alpha z} \right] D_2,\end{aligned}\tag{20}$$

$$\begin{aligned}\tilde{T}_p &= \frac{u_{21}u_{22}}{U} \left[\cosh \sqrt{\lambda_1}z - \cosh \sqrt{\lambda_2}z \right] \tilde{T}_e(0) \\ &+ \frac{1}{U} \left[-u_{12}u_{21} \cosh \sqrt{\lambda_1}z + u_{11}u_{22} \cosh \sqrt{\lambda_2}z \right] \tilde{T}_p(0) \\ &+ u_{21} \left[\frac{\alpha}{\sqrt{\lambda_1}} \sinh \sqrt{\lambda_1}z - \cosh \sqrt{\lambda_1}z + e^{-\alpha z} \right] D_1 \\ &+ u_{22} \left[\frac{\alpha}{\sqrt{\lambda_2}} \sinh \sqrt{\lambda_2}z - \cosh \sqrt{\lambda_2}z + e^{-\alpha z} \right] D_2.\end{aligned}\tag{21}$$

The heat flux carried by electrons and phonons can be similarly expressed in terms of $\tilde{T}_e(0)$, $\tilde{T}_p(0)$, D_1 and D_2 as,

$$\begin{aligned}\tilde{q}_e &= \frac{\kappa_{e,z}}{U} \left[-u_{11}u_{22} \sqrt{\lambda_1} \sinh \sqrt{\lambda_1}z + u_{12}u_{21} \sqrt{\lambda_2} \sinh \sqrt{\lambda_2}z \right] \tilde{T}_e(0) \\ &+ \frac{u_{11}u_{12}\kappa_{e,z}}{U} \left[\sqrt{\lambda_1} \sinh \sqrt{\lambda_1}z - \sqrt{\lambda_2} \sinh \sqrt{\lambda_2}z \right] \tilde{T}_p(0) \\ &+ u_{11}\kappa_{e,z} \left[\sqrt{\lambda_1} \sinh \sqrt{\lambda_1}z - \alpha \cosh \sqrt{\lambda_1}z + \alpha e^{-\alpha z} \right] D_1 \\ &+ u_{12}\kappa_{e,z} \left[\sqrt{\lambda_2} \sinh \sqrt{\lambda_2}z - \alpha \cosh \sqrt{\lambda_2}z + \alpha e^{-\alpha z} \right] D_2,\end{aligned}\tag{22}$$

$$\begin{aligned}\tilde{q}_p &= \frac{u_{21}u_{22}\kappa_{p,z}}{U} \left[-\sqrt{\lambda_1} \sinh \sqrt{\lambda_1}z + \sqrt{\lambda_2} \sinh \sqrt{\lambda_2}z \right] \tilde{T}_e(0) \\ &+ \frac{\kappa_{p,z}}{U} \left[u_{12}u_{21} \sqrt{\lambda_1} \sinh \sqrt{\lambda_1}z - u_{11}u_{22} \sqrt{\lambda_2} \sinh \sqrt{\lambda_2}z \right] \tilde{T}_p(0) \\ &+ u_{21}\kappa_{p,z} \left[\sqrt{\lambda_1} \sinh \sqrt{\lambda_1}z - \alpha \cosh \sqrt{\lambda_1}z + \alpha e^{-\alpha z} \right] D_1 \\ &+ u_{22}\kappa_{p,z} \left[\sqrt{\lambda_2} \sinh \sqrt{\lambda_2}z - \alpha \cosh \sqrt{\lambda_2}z + \alpha e^{-\alpha z} \right] D_2.\end{aligned}\tag{23}$$

The above four equations (Eqs. 20, 21, 22, 23) can be compactly expressed in matrix form,

$$\underbrace{\begin{pmatrix} \tilde{T}_e \\ \tilde{T}_p \\ \tilde{q}_e \\ \tilde{q}_p \end{pmatrix}}_{\mathbf{Q}} = \underbrace{\begin{pmatrix} Q_{11} & Q_{12} & Q_{13} & Q_{14} \\ Q_{21} & Q_{22} & Q_{23} & Q_{24} \\ Q_{31} & Q_{32} & Q_{33} & Q_{34} \\ Q_{41} & Q_{42} & Q_{43} & Q_{44} \end{pmatrix}}_{\mathbf{Q}} \begin{pmatrix} \tilde{T}_e(0) \\ \tilde{T}_p(0) \\ D_1 \\ D_2 \end{pmatrix},\tag{24}$$

where

$$\begin{aligned}
Q_{11} &= \frac{1}{U} \left[u_{11}u_{22} \cosh \sqrt{\lambda_1}z - u_{12}u_{21} \cosh \sqrt{\lambda_2}z \right], \\
Q_{12} &= \frac{u_{11}u_{12}}{U} \left[-\cosh \sqrt{\lambda_1}z + \cosh \sqrt{\lambda_2}z \right], \\
Q_{13} &= u_{11} \left[\frac{\alpha}{\sqrt{\lambda_1}} \sinh \sqrt{\lambda_1}z - \cosh \sqrt{\lambda_1}z + e^{-\alpha z} \right], \\
Q_{14} &= u_{12} \left[\frac{\alpha}{\sqrt{\lambda_2}} \sinh \sqrt{\lambda_2}z - \cosh \sqrt{\lambda_2}z + e^{-\alpha z} \right], \\
Q_{21} &= \frac{u_{21}u_{22}}{U} \left[\cosh \sqrt{\lambda_1}z - \cosh \sqrt{\lambda_2}z \right], \\
Q_{22} &= \frac{1}{U} \left[-u_{12}u_{21} \cosh \sqrt{\lambda_1}z + u_{11}u_{22} \cosh \sqrt{\lambda_2}z \right], \\
Q_{23} &= u_{21} \left[\frac{\alpha}{\sqrt{\lambda_1}} \sinh \sqrt{\lambda_1}z - \cosh \sqrt{\lambda_1}z + e^{-\alpha z} \right], \\
Q_{24} &= u_{22} \left[\frac{\alpha}{\sqrt{\lambda_2}} \sinh \sqrt{\lambda_2}z - \cosh \sqrt{\lambda_2}z + e^{-\alpha z} \right], \\
Q_{31} &= \frac{\kappa_{e,z}}{U} \left[-u_{11}u_{22}\sqrt{\lambda_1} \sinh \sqrt{\lambda_1}z + u_{12}u_{21}\sqrt{\lambda_2} \sinh \sqrt{\lambda_2}z \right], \\
Q_{32} &= \frac{u_{11}u_{12}\kappa_{e,z}}{U} \left[\sqrt{\lambda_1} \sinh \sqrt{\lambda_1}z - \sqrt{\lambda_2} \sinh \sqrt{\lambda_2}z \right], \\
Q_{33} &= u_{11}\kappa_{e,z} \left[\sqrt{\lambda_1} \sinh \sqrt{\lambda_1}z - \alpha \cosh \sqrt{\lambda_1}z + \alpha e^{-\alpha z} \right], \\
Q_{34} &= u_{12}\kappa_{e,z} \left[\sqrt{\lambda_2} \sinh \sqrt{\lambda_2}z - \alpha \cosh \sqrt{\lambda_2}z + \alpha e^{-\alpha z} \right], \\
Q_{41} &= \frac{u_{21}u_{22}\kappa_{p,z}}{U} \left[-\sqrt{\lambda_1} \sinh \sqrt{\lambda_1}z + \sqrt{\lambda_2} \sinh \sqrt{\lambda_2}z \right], \\
Q_{42} &= \frac{\kappa_{p,z}}{U} \left[u_{12}u_{21}\sqrt{\lambda_1} \sinh \sqrt{\lambda_1}z - u_{11}u_{22}\sqrt{\lambda_2} \sinh \sqrt{\lambda_2}z \right], \\
Q_{43} &= u_{21}\kappa_{p,z} \left[\sqrt{\lambda_1} \sinh \sqrt{\lambda_1}z - \alpha \cosh \sqrt{\lambda_1}z + \alpha e^{-\alpha z} \right], \\
Q_{44} &= u_{22}\kappa_{p,z} \left[\sqrt{\lambda_2} \sinh \sqrt{\lambda_2}z - \alpha \cosh \sqrt{\lambda_2}z + \alpha e^{-\alpha z} \right].
\end{aligned} \tag{25}$$

Within each layer other than the top metal transducer layer, the explicit expressions of temperatures write,

$$\begin{aligned}
\begin{pmatrix} \tilde{T}_e \\ \tilde{T}_p \end{pmatrix} &= \mathbf{u} \begin{pmatrix} B_1 e^{-\sqrt{\lambda_1}z} + C_1 e^{\sqrt{\lambda_1}z} \\ B_2 e^{-\sqrt{\lambda_2}z} + C_2 e^{\sqrt{\lambda_2}z} \end{pmatrix} \\
&= \begin{pmatrix} u_{11}B_1 e^{-\sqrt{\lambda_1}z} + u_{11}C_1 e^{\sqrt{\lambda_1}z} + u_{12}B_2 e^{-\sqrt{\lambda_2}z} + u_{12}C_2 e^{\sqrt{\lambda_2}z} \\ u_{21}B_1 e^{-\sqrt{\lambda_1}z} + u_{21}C_1 e^{\sqrt{\lambda_1}z} + u_{22}B_2 e^{-\sqrt{\lambda_2}z} + u_{22}C_2 e^{\sqrt{\lambda_2}z} \end{pmatrix},
\end{aligned} \tag{26}$$

and the heat fluxes are given by,

$$\begin{pmatrix} \tilde{q}_e \\ \tilde{q}_p \end{pmatrix} = \begin{pmatrix} u_{11}B_1\kappa_{e,z}\sqrt{\lambda_1}e^{-\sqrt{\lambda_1}z} - u_{11}C_1\kappa_{e,z}\sqrt{\lambda_1}e^{\sqrt{\lambda_1}z} + u_{12}B_2\kappa_{e,z}\sqrt{\lambda_2}e^{-\sqrt{\lambda_2}z} - u_{12}C_2\kappa_{e,z}\sqrt{\lambda_2}e^{\sqrt{\lambda_2}z} \\ u_{21}B_1\kappa_{p,z}\sqrt{\lambda_1}e^{-\sqrt{\lambda_1}z} - u_{21}C_1\kappa_{p,z}\sqrt{\lambda_1}e^{\sqrt{\lambda_1}z} + u_{22}B_2\kappa_{p,z}\sqrt{\lambda_2}e^{-\sqrt{\lambda_2}z} - u_{22}C_2\kappa_{p,z}\sqrt{\lambda_2}e^{\sqrt{\lambda_2}z} \end{pmatrix}. \tag{27}$$

We reorganize the Eqs. 26 and 27 using matrix algebra,

$$\begin{pmatrix} \tilde{T}_e \\ \tilde{T}_p \\ \tilde{q}_e \\ \tilde{q}_p \end{pmatrix} = \underbrace{\begin{pmatrix} u_{11}e^{-\sqrt{\lambda_1}z} & u_{11}e^{\sqrt{\lambda_1}z} & u_{12}e^{-\sqrt{\lambda_2}z} & u_{12}e^{\sqrt{\lambda_2}z} \\ u_{21}e^{-\sqrt{\lambda_1}z} & u_{21}e^{\sqrt{\lambda_1}z} & u_{22}e^{-\sqrt{\lambda_2}z} & u_{22}e^{\sqrt{\lambda_2}z} \\ u_{11}\kappa_{e,z}\sqrt{\lambda_1}e^{-\sqrt{\lambda_1}z} & -u_{11}\kappa_{e,z}\sqrt{\lambda_1}e^{\sqrt{\lambda_1}z} & u_{12}\kappa_{e,z}\sqrt{\lambda_2}e^{-\sqrt{\lambda_2}z} & -u_{12}\kappa_{e,z}\sqrt{\lambda_2}e^{\sqrt{\lambda_2}z} \\ u_{21}\kappa_{p,z}\sqrt{\lambda_1}e^{-\sqrt{\lambda_1}z} & -u_{21}\kappa_{p,z}\sqrt{\lambda_1}e^{\sqrt{\lambda_1}z} & u_{22}\kappa_{p,z}\sqrt{\lambda_2}e^{-\sqrt{\lambda_2}z} & -u_{22}\kappa_{p,z}\sqrt{\lambda_2}e^{\sqrt{\lambda_2}z} \end{pmatrix}}_{\mathbf{N}} \begin{pmatrix} B_1 \\ C_1 \\ B_2 \\ C_2 \end{pmatrix}. \quad (28)$$

We also note that the newly defined matrix $\mathbf{N}$ can be factorized into the product of the following two matrices,

$$\mathbf{N} = \begin{pmatrix} u_{11} & u_{11} & u_{12} & u_{12} \\ u_{21} & u_{21} & u_{22} & u_{22} \\ u_{11}\kappa_{e,z}\sqrt{\lambda_1} & -u_{11}\kappa_{e,z}\sqrt{\lambda_1} & u_{12}\kappa_{e,z}\sqrt{\lambda_2} & -u_{12}\kappa_{e,z}\sqrt{\lambda_2} \\ u_{21}\kappa_{p,z}\sqrt{\lambda_1} & -u_{21}\kappa_{p,z}\sqrt{\lambda_1} & u_{22}\kappa_{p,z}\sqrt{\lambda_2} & -u_{22}\kappa_{p,z}\sqrt{\lambda_2} \end{pmatrix} \underbrace{\begin{pmatrix} e^{-\sqrt{\lambda_1}z} & & & \\ & e^{\sqrt{\lambda_1}z} & & \\ & & e^{-\sqrt{\lambda_2}z} & \\ & & & e^{\sqrt{\lambda_2}z} \end{pmatrix}}_{\mathbf{E}} \quad (29)$$

$= \mathbf{N}(z=0)\mathbf{E},$

where

$$\mathbf{E} = \cosh \sqrt{\lambda_2}z \begin{pmatrix} \frac{\cosh \sqrt{\lambda_1}z}{\cosh \sqrt{\lambda_2}z} (1 - \tanh \sqrt{\lambda_1}z) & & & \\ & \frac{\cosh \sqrt{\lambda_1}z}{\cosh \sqrt{\lambda_2}z} (1 + \tanh \sqrt{\lambda_1}z) & & \\ & & 1 - \tanh \sqrt{\lambda_2}z & \\ & & & 1 + \tanh \sqrt{\lambda_2}z \end{pmatrix}. \quad (30)$$

In particular, at surface $z = 0$, we have,

$$\begin{pmatrix} \tilde{T}_e(0) \\ \tilde{T}_p(0) \\ \tilde{q}_e(0) \\ \tilde{q}_p(0) \end{pmatrix} = \mathbf{N}(z=0) \begin{pmatrix} B_1 \\ C_1 \\ B_2 \\ C_2 \end{pmatrix}, \quad (31)$$

which can alternatively be written as,

$$\begin{pmatrix} B_1 \\ C_1 \\ B_2 \\ C_2 \end{pmatrix} = [\mathbf{N}(z=0)]^{-1} \begin{pmatrix} \tilde{T}_e(0) \\ \tilde{T}_p(0) \\ \tilde{q}_e(0) \\ \tilde{q}_p(0) \end{pmatrix}. \quad (32)$$

Hence, for any layer, the temperature and heat flux at the bottom surface can be related to those at the top surface via,

$$\begin{pmatrix} \tilde{T}_e(l) \\ \tilde{T}_p(l) \\ \tilde{q}_e(l) \\ \tilde{q}_p(l) \end{pmatrix} = \mathbf{N}(z=l) [\mathbf{N}(z=0)]^{-1} \begin{pmatrix} \tilde{T}_e(0) \\ \tilde{T}_p(0) \\ \tilde{q}_e(0) \\ \tilde{q}_p(0) \end{pmatrix}, \quad (33)$$

where l is the thickness of the layer.

Across the interface between neighboring layers, we have,

$$\begin{pmatrix} \tilde{T}'_e(0) \\ \tilde{T}'_p(0) \\ \tilde{q}'_e(0) \\ \tilde{q}'_p(0) \end{pmatrix} = \underbrace{\begin{pmatrix} 1 & 0 & -1/G_{ee} & 0 \\ 0 & 1 & 0 & -1/G_{pp} \\ -G_{pe} & G_{pe} & 1 + \frac{G_{pe}}{G_{ee}} & 0 \\ G_{ep} & -G_{ep} & 0 & 1 + \frac{G_{ep}}{G_{pp}} \end{pmatrix}}_{\mathbf{G}} \begin{pmatrix} \tilde{T}_e(l) \\ \tilde{T}_p(l) \\ \tilde{q}_e(l) \\ \tilde{q}_p(l) \end{pmatrix}, \quad (34)$$

where the ' symbol indicates the layer immediately beneath the current one, G_{ee} is the electronic thermal boundary conductance, and G_{pp} is the phonon thermal boundary conductance. Specifically, G_{ep} and G_{pe} refer to thermal conductances arising from the non-local electron-phonon interaction across the interface (electrons in the current layer directly interact with phonons in the next layer, and phonons in the current layer directly interact with electrons in the next layer). Their existence has never been experimentally confirmed, and any such contribution is weak. Therefore, we set $G_{ep} = G_{pe} = 0$ in this work.

As a result, we are able to relate the temperature and heat flux at the bottom surface of the last layer to the temperature and heat input at the top surface of the first layer via,

$$\begin{pmatrix} \tilde{T}_{b,e} \\ \tilde{T}_{b,p} \\ \tilde{q}_{b,e} \\ \tilde{q}_{b,p} \end{pmatrix} = \mathbf{T} \begin{pmatrix} \tilde{T}_e(0) \\ \tilde{T}_p(0) \\ D_1 \\ D_2 \end{pmatrix}, \quad (35)$$

where the matrix $\mathbf{T}$ is defined by the following (ordered) product of matrices,

$$\mathbf{T} = \mathbf{N}_n(l_n) [\mathbf{N}_n(0)]^{-1} \mathbf{G}_{n,n-1} \cdots \mathbf{G}_{4,3} \mathbf{N}_3(l_3) [\mathbf{N}_3(0)]^{-1} \mathbf{G}_{3,2} \mathbf{N}_2(l_2) [\mathbf{N}_2(0)]^{-1} \mathbf{G}_{2,1} \mathbf{Q}(l_1). \quad (36)$$

Here, the subscript n indicates the n -th layer from the top.

At the bottom surface of the last layer, i.e., the substrate, the heat fluxes are zero (adiabatic boundary condition) such that $\tilde{q}_{b,e} = \tilde{q}_{b,p} = 0$. This leads to,

$$T_{31}\tilde{T}_e(0) + T_{32}\tilde{T}_p(0) + T_{33}D_1 + T_{34}D_2 = 0, \quad (37)$$

$$T_{41}\tilde{T}_e(0) + T_{42}\tilde{T}_p(0) + T_{43}D_1 + T_{44}D_2 = 0. \quad (38)$$

Note that the thermoreflectance signal is predominantly contributed by the surface phonon temperature instead of the surface electron temperature, as pointed by prior work[1]. Hence, by further eliminating $\tilde{T}_e(0)$, it follows that,

$$\begin{aligned} \tilde{T}_p(0) &= \frac{T_{31}T_{43} - T_{33}T_{41}}{T_{32}T_{41} - T_{31}T_{42}} D_1 + \frac{T_{31}T_{44} - T_{34}T_{41}}{T_{32}T_{41} - T_{31}T_{42}} D_2 \\ &= \frac{\tilde{P}(z=0)}{\kappa_{e,z}} \left[\frac{T_{31}T_{43} - T_{33}T_{41}}{T_{32}T_{41} - T_{31}T_{42}} \frac{v_{11}}{\lambda_1 - \alpha^2} + \frac{T_{31}T_{44} - T_{34}T_{41}}{T_{32}T_{41} - T_{31}T_{42}} \frac{v_{21}}{\lambda_2 - \alpha^2} \right]. \end{aligned} \quad (39)$$

We now have the Green's function, defined as the ratio between the surface temperature of phonons in the metal layer and the pump heat power,

$$\begin{aligned} \tilde{G} &= \frac{\tilde{T}_p(0)}{\tilde{P}(0)} \\ &= \frac{1}{\kappa_{e,z}} \left[\frac{T_{31}T_{43} - T_{33}T_{41}}{T_{32}T_{41} - T_{31}T_{42}} \frac{v_{11}}{\lambda_1 - \alpha^2} + \frac{T_{31}T_{44} - T_{34}T_{41}}{T_{32}T_{41} - T_{31}T_{42}} \frac{v_{21}}{\lambda_2 - \alpha^2} \right]. \end{aligned} \quad (40)$$

Finally, after applying the inverse Fourier transform, we obtain the measured thermoreflectance signal,

$$I \approx \frac{\text{CTR}}{(2\pi)^2} \iint \tilde{T}_p(0) e^{-q_x^2 R_x^2/4} e^{-q_y^2 R_y^2/4} e^{-iq_x x_0} e^{-iq_y y_0} d^2 \mathbf{q}_{\parallel}, \quad (41)$$

where $R_x = \sqrt{(\sigma_x^2 + \sigma_x'^2)/2}$, $R_y = \sqrt{(\sigma_y^2 + \sigma_y'^2)/2}$ are the effective radii, σ'_x and σ'_y are the probe beam radii along the x and y directions, x_o and y_o are the beam offset distances between pump and probe centers, and CTR is the coefficient of thermorefectance of the metal. When there is no beam offset and the beam profiles are isotropic ($R_x = R_y = R$), Eq. 41 reduces to,

$$\begin{aligned} I &= \frac{\text{CTR}}{2\pi} \int \tilde{T}_p(0) e^{-q_{\parallel}^2 R^2/4} q_{\parallel} dq_{\parallel} \\ &= \frac{\text{CTR}}{2\pi} \int \tilde{G} \tilde{P}(0) e^{-q_{\parallel}^2 R^2/4} q_{\parallel} dq_{\parallel}. \end{aligned} \quad (42)$$

2 Case study: spin-crossover complex coated with a gold transducer

We consider the case of Au transducer on a spin-crossover complex (simplified hereafter as SCO). Since electrons contributed negligibly to heat conduction due to the small electrical conductivity, we consider only electrons and phonons in Au, and phonons in SCO.

Inside SCO, the solution for phonon temperature takes the form,

$$\tilde{T}_{\text{SCO},p} = B_1 e^{-\sqrt{\lambda}z} + C_1 e^{\sqrt{\lambda}z}, \quad (43)$$

where $\lambda = \frac{C_p i\omega + \kappa_{\text{SCO},\parallel} q_{\parallel}^2}{\kappa_{\text{SCO},p}}$. The phonon heat flux is given by,

$$\tilde{q}_{\text{SCO},p} = \kappa_{\text{SCO},p} \sqrt{\lambda} B_1 e^{-\sqrt{\lambda}z} - \kappa_{\text{SCO},p} \sqrt{\lambda} C_1 e^{\sqrt{\lambda}z}. \quad (44)$$

In matrix form, the above two equations can be expressed as,

$$\begin{aligned} \begin{pmatrix} \tilde{T}_{\text{SCO},p}(z) \\ \tilde{q}_{\text{SCO},p}(z) \end{pmatrix} &= \begin{pmatrix} e^{-\sqrt{\lambda}z} & e^{\sqrt{\lambda}z} \\ \kappa_{\text{SCO},p} \sqrt{\lambda} e^{-\sqrt{\lambda}z} & -\kappa_{\text{SCO},p} \sqrt{\lambda} e^{\sqrt{\lambda}z} \end{pmatrix} \begin{pmatrix} B_1 \\ C_1 \end{pmatrix} \\ &= \begin{pmatrix} e^{-\sqrt{\lambda}z} & e^{\sqrt{\lambda}z} \\ \kappa_{\text{SCO},p} \sqrt{\lambda} e^{-\sqrt{\lambda}z} & -\kappa_{\text{SCO},p} \sqrt{\lambda} e^{\sqrt{\lambda}z} \end{pmatrix} \begin{pmatrix} 1 & 1 \\ \kappa_{\text{SCO},p} \sqrt{\lambda} & -\kappa_{\text{SCO},p} \sqrt{\lambda} \end{pmatrix}^{-1} \begin{pmatrix} \tilde{T}_{\text{SCO},p}(0) \\ \tilde{q}_{\text{SCO},p}(0) \end{pmatrix} \\ &= \begin{pmatrix} \cosh \sqrt{\lambda}z & -\frac{\sinh \sqrt{\lambda}z}{\kappa_{\text{SCO},p} \sqrt{\lambda}} \\ -\kappa_{\text{SCO},p} \sqrt{\lambda} \sinh \sqrt{\lambda}z & \cosh \sqrt{\lambda}z \end{pmatrix} \begin{pmatrix} \tilde{T}_{\text{SCO},p}(0) \\ \tilde{q}_{\text{SCO},p}(0) \end{pmatrix}. \end{aligned} \quad (45)$$

Therefore, at the bottom side of the SCO layer,

$$\begin{pmatrix} \tilde{T}_{\text{SCO},p}(L) \\ \tilde{q}_{\text{SCO},p}(L) \end{pmatrix} = \underbrace{\begin{pmatrix} \cosh \sqrt{\lambda}L & -\frac{\sinh \sqrt{\lambda}L}{\kappa_{\text{SCO},p} \sqrt{\lambda}} \\ -\kappa_{\text{SCO},p} \sqrt{\lambda} \sinh \sqrt{\lambda}L & \cosh \sqrt{\lambda}L \end{pmatrix}}_{\mathbf{F}} \begin{pmatrix} \tilde{T}_{\text{SCO},p}(0) \\ \tilde{q}_{\text{SCO},p}(0) \end{pmatrix}, \quad (46)$$

where L is the thickness of the SCO complex sample.

We deposit a 93-nm Au transducer with 2-nm Ti adhesion layer between Au and SCO. Since the Ti layer is extremely thin compared to the Au layer, we effectively lumped the thermal resistances associated with the Ti layer into that of the Au/SCO interface. We have also verified that such simplification results in negligible discrepancy compared to the full model including the Ti layer (with both electrons and phonons). Accordingly, at the Au/SCO interface, we have,

$$\begin{pmatrix} \tilde{T}_{\text{SCO},p}(0) \\ \tilde{q}_{\text{SCO},p}(0) \end{pmatrix} = \underbrace{\begin{pmatrix} \frac{G_{ep}}{G_{ep}+G_{pp}} & \frac{G_{pp}}{G_{ep}+G_{pp}} & -(G_{ep}+G_{pp})^{-1} & -(G_{ep}+G_{pp})^{-1} \\ 0 & 0 & 1 & 1 \end{pmatrix}}_{\mathbf{G}} \begin{pmatrix} \tilde{T}_{\text{Au},e}(l) \\ \tilde{T}_{\text{Au},p}(l) \\ \tilde{q}_{\text{Au},e}(l) \\ \tilde{q}_{\text{Au},p}(l) \end{pmatrix}, \quad (47)$$

where l is the thickness of the Au layer and the temperature and heat flux at the bottom side of the first layer is given by Eq. 24. Note that the nonlocal electron-phonon coupling across interface gives rise to the heat current,

$$\tilde{q}_{\text{Au},e}(l) = G_{ep} [\tilde{T}_{\text{Au},e}(l) - \tilde{T}_{\text{SCO},p}(0)], \quad (48)$$

which also serves as a boundary condition. The phonon heat flux at the Au side of the interface is,

$$\tilde{q}_{\text{Au},p}(l) = G_{pp} [\tilde{T}_{\text{Au},p}(l) - \tilde{T}_{\text{SCO},p}(0)]. \quad (49)$$

Eventually, we can relate the bottom-side temperature of the SCO to the top-side temperature and heating power at the Au surface via,

$$\begin{pmatrix} \tilde{T}_{\text{SCO},p}(L) \\ \tilde{q}_{\text{SCO},p}(L) \end{pmatrix} = \underbrace{\begin{pmatrix} T_{11} & T_{12} & T_{13} & T_{14} \\ T_{21} & T_{22} & T_{23} & T_{24} \end{pmatrix}}_{\mathbf{T}=\mathbf{FGQ}} \begin{pmatrix} \tilde{T}_{\text{Au},e}(0) \\ \tilde{T}_{\text{Au},p}(0) \\ D_1 \\ D_2 \end{pmatrix}. \quad (50)$$

Consider the adiabatic boundary condition on the bottom side of SCO, $\tilde{q}_{\text{SCO},p}(L) = 0$. We have,

$$T_{21}\tilde{T}_{\text{Au},e}(0) + T_{22}\tilde{T}_{\text{Au},p}(0) + T_{23}D_1 + T_{24}D_2 = 0. \quad (51)$$

In addition, Eq. 48 results in the following expression,

$$X_1\tilde{T}_{\text{Au},e}(0) + X_2\tilde{T}_{\text{Au},p}(0) + X_3D_1 + X_4D_2 = 0, \quad (52)$$

where $X_i = Q_{3i} - G_{ep}Q_{1i} + G_{ep}(GQ)_{1i}$, forming a vector $\mathbf{X} = (X_1, X_2, X_3, X_4)^T$, and $(GQ)_{1i}$ is the elements of the first row of the matrix $\mathbf{GQ}$.

Hence, by eliminating $\tilde{T}_{\text{Au},e}(0)$ in Eq. 51 and Eq. 52, we arrive at,

$$\begin{aligned} \tilde{T}_{\text{Au},p}(0) &= -\frac{1}{X_1T_{22} - X_2T_{21}} [(X_1T_{23} - X_3T_{21})D_1 + (X_1T_{24} - X_4T_{21})D_2] \\ &= -\frac{\tilde{P}(z=0)}{\kappa_{e,z}} \left[\frac{X_1T_{23} - X_3T_{21}}{X_1T_{22} - X_2T_{21}} \frac{v_{11}}{\lambda_1 - \alpha^2} + \frac{X_1T_{24} - X_4T_{21}}{X_1T_{22} - X_2T_{21}} \frac{v_{21}}{\lambda_2 - \alpha^2} \right]. \end{aligned} \quad (53)$$

Then, the expression for $\tilde{T}_{\text{Au},e}(0)$ follows,

$$\begin{aligned} \tilde{T}_{\text{Au},e}(0) &= \frac{1}{X_1T_{22} - X_2T_{21}} [(X_2T_{23} - X_3T_{22})D_1 + (X_2T_{24} - X_4T_{22})D_2] \\ &= \frac{\tilde{P}(z=0)}{\kappa_{e,z}} \left[\frac{X_2T_{23} - X_3T_{22}}{X_1T_{22} - X_2T_{21}} \frac{v_{11}}{\lambda_1 - \alpha^2} + \frac{X_2T_{24} - X_4T_{22}}{X_1T_{22} - X_2T_{21}} \frac{v_{21}}{\lambda_2 - \alpha^2} \right]. \end{aligned} \quad (54)$$

We can immediately know the coefficients B_1 and B_2 introduced in Eqs. 18 and 19 by plugging the above expressions (Eqs. 53 and 54) into those equations. This, in turn, allows us to plot the electron and phonon temperatures within the metal layer. Additionally, we identify that the Green's function for surface phonon temperature is given by,

$$\begin{aligned} \tilde{G} &= \frac{\tilde{T}_{\text{Au},p}(0)}{\tilde{P}(z=0)} \\ &= -\frac{1}{\kappa_{e,z}(X_1T_{22} - X_2T_{21})} \\ &\quad \times \left[(X_1T_{23} - X_3T_{21}) \frac{v_{11}}{\lambda_1 - \alpha^2} + (X_1T_{24} - X_4T_{21}) \frac{v_{21}}{\lambda_2 - \alpha^2} \right]. \end{aligned} \quad (55)$$

With the Green's function, one can evaluate the temperature according to Eq. 41.

3 Numerical implementation

The elements of matrix $\mathbf{Q}$ can vary in a wide range of order of magnitudes, causing challenges in numerical stability. To reduce the computational errors, we rescale some elements of $\mathbf{Q}$ such that the Eq. 24 becomes,

$$\begin{pmatrix} \tilde{T}_e \\ \tilde{T}_p \\ s\tilde{q}_e \\ s\tilde{q}_p \end{pmatrix} = \underbrace{\begin{pmatrix} Q_{11} & Q_{12} & Q_{13} & Q_{14} \\ Q_{21} & Q_{22} & Q_{23} & Q_{24} \\ sQ_{31} & sQ_{32} & sQ_{33} & sQ_{34} \\ sQ_{41} & sQ_{42} & sQ_{43} & sQ_{44} \end{pmatrix}}_{\mathbf{Q}'} \begin{pmatrix} \tilde{T}_e(0) \\ \tilde{T}_p(0) \\ D_1 \\ D_2 \end{pmatrix}. \quad (56)$$

A good choice of the scaling factor is $s = (|\sqrt{\lambda_2}|\kappa_{p,z})^{-1}$. Note that we denote the rescaled matrix $\mathbf{Q}'$.

Accordingly, the Eq. 47 that describes the interface becomes,

$$\begin{pmatrix} \tilde{T}_{\text{SCO},p}(0) \\ s\tilde{q}_{\text{SCO},p}(0) \end{pmatrix} = \underbrace{\begin{pmatrix} \frac{G_{ep}}{G_{ep}+G_{pp}} & \frac{G_{pp}}{G_{ep}+G_{pp}} & -(G_{ep}+G_{pp})^{-1}s^{-1} & -(G_{ep}+G_{pp})^{-1}s^{-1} \\ 0 & 0 & 1 & 1 \end{pmatrix}}_{\mathbf{G}'} \begin{pmatrix} \tilde{T}_{\text{Au},e}(l) \\ \tilde{T}_{\text{Au},p}(l) \\ s\tilde{q}_{\text{Au},e}(l) \\ s\tilde{q}_{\text{Au},p}(l) \end{pmatrix}. \quad (57)$$

As for the SCO layer, Eq. 46 now writes,

$$\begin{pmatrix} \tilde{T}_{\text{SCO},p}(L) \\ \tilde{q}_{\text{SCO},p}(L) \end{pmatrix} = \underbrace{\begin{pmatrix} \cosh \sqrt{\lambda}L & -s^{-1} \frac{\sinh \sqrt{\lambda}L}{\kappa_{\text{SCO},p} \sqrt{\lambda}} \\ -\kappa_{\text{SCO},p} \sqrt{\lambda} \sinh \sqrt{\lambda}L & s^{-1} \cosh \sqrt{\lambda}L \end{pmatrix}}_{\mathbf{F}'} \begin{pmatrix} \tilde{T}_{\text{SCO},p}(0) \\ s\tilde{q}_{\text{SCO},p}(0) \end{pmatrix}. \quad (58)$$

Correspondingly, the elements of vector $\mathbf{X}$ now become,

$$X_i = s^{-1}Q'_{3i} - G_{ep}Q'_{1i} + G_{ep}(G'Q')_{1i}. \quad (59)$$

4 Note on the non-local electron-phonon interactions across the interface

In the above derivation, we introduced the thermal conductance arising from the non-local electron-phonon interaction across the interface, G_{ep} . Although we have set it to zero in our numerical calculations, it is worth discussing how this new channel impacts the overall interfacial thermal conductance, as it has not been addressed in any thermoreflectance studies.

In the conventional picture of interfacial thermal transport across a metal-insulator interface, the interfacial thermal conductance is given by[2],

$$G = \frac{1}{\frac{1}{G'_{ep}} + \frac{1}{G_{pp}}} = \frac{G'_{ep}G_{pp}}{G'_{ep} + G_{pp}}, \quad (60)$$

where G_{pp} is the conductance for the phonon channel and $G'_{ep} = \sqrt{g\kappa_p}$ is the effective conductance for electron-phonon coupling in the metal side. It is clear that the resistances for such two channels are in series. However, since we have introduced a separate, non-local electron-phonon coupling channel across the interface, G_{ep} , we need to rederive the effective interfacial thermal conductance.

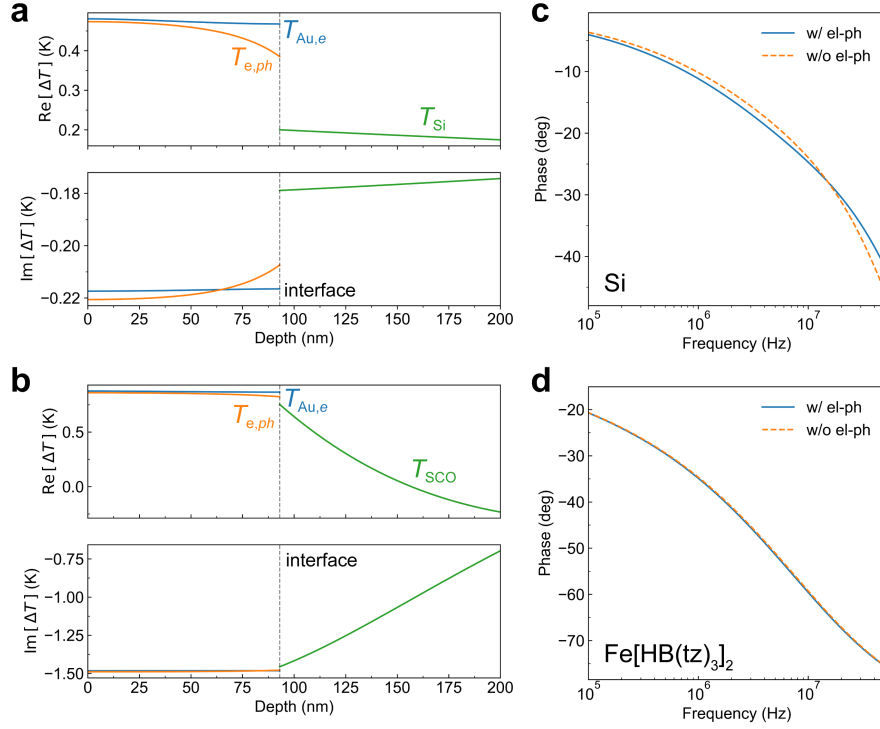


Figure 1: Effect of electron-phonon coupling on FDTR signals. a,b, The electron and phonon temperature profiles in the Au transducer layer, and the phonon temperatures in the non-metallic substrates: Si (a) and Fe[HB(tz)₃]₂ (b). The vertical dashed lines indicate the location of the interface. c,d, Calculated phase of thermorefectance at 295 K for Au/Si (c) and Au/Fe[HB(tz)₃]₂ (d) samples, with and without considering electron-phonon coupling in the metallic layer. Parameters used: $\kappa_{\text{Au,e}} = 165 \text{ W m}^{-1}\text{K}^{-1}$, $\kappa_{\text{Au,p}} = 10 \text{ W m}^{-1}\text{K}^{-1}$, $C_{\text{Au,e}} = 2 \times 10^4 \text{ J m}^{-3}\text{K}^{-1}$, $C_{\text{Au,p}} = 2.48 \times 10^6 \text{ J m}^{-3}\text{K}^{-1}$, $g = 2 \times 10^{16} \text{ W m}^{-3}\text{K}^{-1}$, and $P_0 = 0.6 \text{ mW}$.

At steady-state, the electron and phonon temperatures in the metal layer satisfy,

$$\begin{aligned}\kappa_e \frac{\partial^2 T_e}{\partial z^2} - g(T_e - T_p) &= 0, \\ \kappa_p \frac{\partial^2 T_p}{\partial z^2} + g(T_e - T_p) &= 0,\end{aligned}\tag{61}$$

which leads to,

$$\frac{\partial^2 \theta}{\partial z^2} - \frac{\theta}{l^2} = 0,\tag{62}$$

where $\theta = T_e - T_p$ and $\frac{1}{l^2} = g \frac{\kappa_e + \kappa_p}{\kappa_e \kappa_p}$. The solution for θ takes the form,

$$\theta = B e^{-z/l}.\tag{63}$$

This allows us to express the electron temperature as,

$$T_e = \frac{g l^2}{\kappa_e} B e^{-z/l} + C z + D\tag{64}$$

and the phonon temperature as,

$$T_p = \left(\frac{g l^2}{\kappa_e} - 1 \right) B e^{-z/l} + C z + D.\tag{65}$$

At the interface, we have the following boundary conditions,

$$-\kappa_e \frac{\partial T_e}{\partial z} \Big|_{z=0} = -G_{ep} [T_e(0) - T(0)],\tag{66}$$

$$-\kappa_p \frac{\partial T_p}{\partial z} \Big|_{z=0} = -G_{pp} [T_p(0) - T(0)],\tag{67}$$

$$-\kappa_e \frac{\partial T_e}{\partial z} \Big|_{z=0} - \kappa_p \frac{\partial T_p}{\partial z} \Big|_{z=0} = -\kappa \frac{\partial T}{\partial z} \Big|_{z=0},\tag{68}$$

where $T = Az$ is the temperature profile on the semiconductor side.

We can then express C and D in terms of B ,

$$C = \frac{G_{ep} G_{pp} + \frac{1}{l} \frac{\kappa_e \kappa_p}{\kappa_e + \kappa_p} (G_{ep} + G_{pp})}{\kappa_e G_{pp} - \kappa_p G_{ep}} B,\tag{69}$$

$$D = \left[-\frac{1}{l} \frac{\kappa_e \kappa_p}{\kappa_e + \kappa_p} \frac{1}{G_{ep}} - \frac{\kappa_p}{\kappa_e + \kappa_p} + \frac{\kappa_e}{G_{ep}} \frac{G_{ep} G_{pp} + \frac{1}{l} \frac{\kappa_e \kappa_p}{\kappa_e + \kappa_p} (G_{ep} + G_{pp})}{\kappa_e G_{pp} - \kappa_p G_{ep}} \right] B.\tag{70}$$

For strong electron-phonon coupling, the temperature profiles for electron and phonon overlaps, and are both linear. At the interface, the temperature discontinuity is just D . In this case, the *effective thermal conductance* can be defined by,

$$G = \frac{-\kappa \frac{\partial T}{\partial z} \Big|_{z=0}}{-D} = \left(1 + \frac{\kappa_p}{\kappa_e} \right) \frac{\left(1 + \frac{\kappa_p}{\kappa_e} \right) G_{ep} G_{pp} + \sqrt{g \left(1 + \frac{\kappa_p}{\kappa_e} \right) \kappa_p (G_{ep} + G_{pp})}}{\sqrt{g \left(1 + \frac{\kappa_p}{\kappa_e} \right) \kappa_p \left(1 + \frac{\kappa_p}{\kappa_e} \right) + \left(\frac{\kappa_p}{\kappa_e} \right)^2 G_{ep} + G_{pp}}}.\tag{71}$$

In metals such as Au, we have $\kappa_p \ll \kappa_e$. In addition, we expect that G_{ep} is much smaller than G_{pp} . Therefore, we can simplify the effective thermal conductance as,

$$\begin{aligned} G &= \frac{G_{ep}G_{pp} + G'_{ep}(G_{ep} + G_{pp})}{G'_{ep} + G_{pp}} \\ &= G_{ep} + \frac{G'_{ep}G_{pp}}{G'_{ep} + G_{pp}}. \end{aligned} \tag{72}$$

The thermal resistance associated with the electron-phonon interaction on the metal side and the thermal resistance for phonon transport across interface are in series, while this combined thermal resistance and the thermal resistance for the non-local electron-phonon interaction across the interface are in *parallel*.

References

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